

Importance of N₂H_x Reaction Pathways in Modelling of NH₃/H₂/Air Flame Chemistry

Aahil Ashraf Urothiyil¹, Abe Shunnosuke¹, Matsuda Dai¹, Kitagawa Toshiaki¹, Ekenechukwu Chijioke Okafor¹

¹Department of Mechanical Engineering, Kyushu University
7-4-4, Motoooka, Nishi-ku, Fukuoka, 819-0395, Japan

¹Student

Street, City, State, Zip-code, Country

ABSTRACT

To investigate the importance of N₂H_x reaction pathways in modelling NH₃/H₂/Air flames, the LBV of the mixture was modelled using an existing reaction mechanism. This mechanism reproduced LBV of NH₃/Air and NH₃/CH₄/Air flames accurately as the mechanism was optimized and validated for these fuel blends however, LBV of NH₃/H₂/Air were under-predicted. Comparing reaction pathways and LBV sensitivities of other mechanisms that were optimized for NH₃/H₂/Air, the importance of N₂H_x reaction pathways was established. A present mechanism was developed and then validated with data from literature.

1. Introduction

Ammonia has gained interest as a carbon free fuel for combustion systems and the blending of ammonia with other faster burning fuel, such as methane and hydrogen, is a means considered for improving the combustion characteristics of ammonia mixtures. To further study the phenomena observed with the combustion of ammonia and its blends under varying conditions, numerical simulations with chemical kinetics mechanisms is necessary. These choice of mechanisms determine the accuracy of the numerically simulated flames. Reaction mechanisms that are able to accurately predict parameters of the flame, such as laminar burning velocity (LBV), under varying conditions, like rich/lean equivalence ratio and varying fuel compositions, is desired.

Many chemical kinetic mechanisms that model NH₃/Air flames or specific blends of ammonia exist. Most of these mechanisms were developed, optimized and validated for targeted mixtures, specific set of flame characteristics and select experimental conditions. Okafor Reduced mechanism, for instance, ...

The aim of this study is to develop a reaction mechanism to accurately model flames for ammonia and ammonia blends — namely NH₃/H₂/Air and CH₄/NH₃/Air flames. This mechanism prioritizes LBV, a critical flame characteristic that dictates fuel compatibility for a wide range of practical applications. Most existing reaction mechanisms from literature are only able to reproduce some fuels' LBV accurately under select pressure, initial temperature and equivalence ratios. Okafor Reduced mechanism, for example, is designed to predict CH₄/NH₃/Air's LBV. This mechanism also exhibits high reproducibility for LBVs of NH₃/Air flames and H₂/Air flames however underpredicts LBV of NH₃/H₂/Air flames. In this study, the reason for the underprediction was studied and a mechanism was developed around the findings.

2. Developing Reaction Mechanism

The new reaction mechanism developed in this paper was based on the Okafor Reduced mechanism. Okafor reduced mechanism was selected because this mechanism has been optimized and validated for LBV prediction of NH₃/Air and NH₃/CH₄/Air flames. This is true for

varying equivalence ratios, pressures and, in the case of NH₃/CH₄/Air flames, varying methane concentrations. This mechanism also models the LBV of H₂/Air flames satisfactorily for most of these conditions. This mechanism performs poorly when it is used to predict LBV of NH₃/H₂/Air flames at hydrogen concentrations greater than 0.3. The mechanism's failure to predict the LBV for this specific blend, despite accurately reproducing the individual components of NH₃/H₂/Air flames, motivated the current investigation. Correcting this discrepancy brings this mechanism to fulfil the aim of this study.

[Figure: lbv NH₃, H₂, NH₃/H₂]

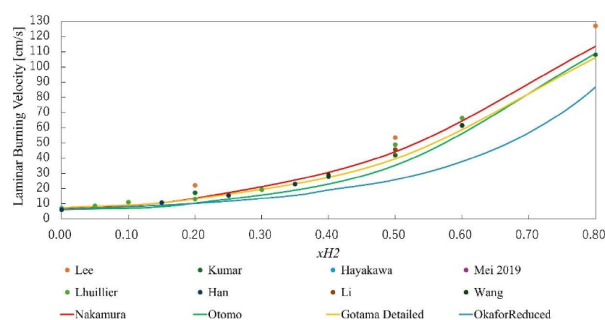


Fig. 1 Experimental and numerical laminar burning velocity of NH₃/H₂/Air flames under stoichiometric condition at various hydrogen concentrations

Nakamura Mech [1] is one of the mechanisms studied to develop the Present Mech and the mechanism on NH₃/Air flames. This mechanism was selected because Nakamura et al. identifies the important role N₂H_x species play in the ignition of NH₃/Air flames, especially at lower temperatures [2]. Hence, the mechanism has a detailed description of how N₂H_x species interact with intermediate species and radicals to control the reactivity of ammonia. Moreover, Nakamura Mech's predictions for NO and N₂O molar fractions, downstream of the reaction zone, laminar burning velocity and ignition delay time agree satisfactorily with data from literature.

Another mechanism that was studied to develop the Present Mech was Gotama Mech [3]. This mechanism was developed to model NH₃/H₂/Air flames for laminar burning velocity. Gotama et al. re-emphasizes the importance of N₂H_x reaction pathways in the combustion of flames containing ammonia. The author outlines an

H2 recovery mechanism in which NH3 and H2 kinetics are coupled to regulate H radical and H2 concentrations [4]. Using this mechanism, Gotama Mech is able to accurately model NH3/H2/Air flames at varying hydrogen concentrations. Implementing the H2 recovery mechanism could improve Present Mech's NHx and N2Hx kinetics for flames with different hydrogen concentrations.

2.1 Methodology

Numerical simulations for this study is conducted using ANSYS Chemkin-PRO 2019 R3 and ANSYS Chemkin 2025 R2. The "Premixed Laminar Flame Speed Calculation" model was used and the initial conditions kept constant for all the simulations were the pressure at 1.0 atm, temperature at 298.0 K, and equivalence ratio at 1.0. To study the effects of hydrogen addition to the blend, the simulations were run with fuel composed of xH2=0.5 and xNH3=0.5. xH2=0.5 was selected to compare these mechanisms as Nakamura Mech and Gotama Mech estimate LBV with reasonable accuracy at these conditions and Okafor Reduced Mech deviates significantly from the experimental LBV at this ratio. The computational domains was set to 10 cm and the solver properties GRAD and CURV were set to 0.01. Multicomponent transport and thermal diffusion — Soret effect — was enabled.

After simulating the flames with the settings mentioned, reaction pathways were created using integrated rate of production (IROP) of each elementary reaction in the mechanisms. Rate of production (ROP) is evaluated at each space step in the computational domain and proves to be a difficult parameter to use to compare elementary reactions of different mechanisms because different reactions' ROPs peak along different points in the domain. To address this limitation, ROP of each elementary reaction is integrated along its domain (1) resulting in one number to represent an elementary reaction's ROP across the domain.

$$IROP = \int_0^L \dot{\omega}_{R,i} dx \quad (1)$$

For Okafor Reduced Mech, IROPs are then normalized with respect to the IROP of the OH→H2O pathway, the reaction pathway with the highest. The normalized IROPs are used to draw the reaction pathways and the normalized IROP value for each pathway determine the thickness of the line. Gotama and Nakamura Mechs will use "Relative Size", the IROP of the pathway relative to Okafor Reduced Mech, to determine the line thicknesses in their reaction mechanisms.

Then the normalized sensitivity of burning velocity, and mole fractions of species of interest, with respect to the rate constants of the elementary reactions were evaluated for Okafor Reduced Mech and the Present Mech. The species of interest were H, OH, NH, NH2, N2H2, NNH and N2H3 as these intermediate species were identified to be significant in NH3/H2 flame chemistry. The sensitivity coefficients represent how the A-factors of each elementary reaction impacts the parameters — LBV or the species mentioned. Comparing the sensitivity analysis plots give insights into which elementary reactions are significant with respect to each parameter and how they affect the parameters in the updated Present Mech.

2.2 Findings from Comparing Mechanisms

[Figures reaction pathways]

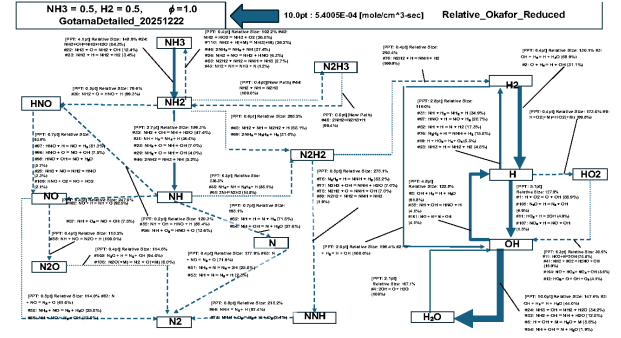


Fig. 2 Reaction pathway of NH3/H2/Air flame at xH2=0.5 and equivalence ratio of 1.0 using Gotama Detailed Mechanism

With Okafor Reduced Mech and Gotama Mech, NH3/H2/Air flames were numerical simulated under conditions and setup outlined in the Methodology. IROPs for both mechanisms were calculated and was used to construct Fig 2, a reaction pathway comparing each Gotama Mech pathway to the corresponding Okafor reduced Mech pathway. The most notable pathway in Fig 2 is NH2↔N2H3 which is only significant in Gotama Mech. The forward reaction, the formation of N2H3, is given by NH2+NH=N2H3 (R1) and the reverse, the consumption of N2H3, is given by 2N2H3=N2H3+H (R2). These reactions affect the concentrations of NH and NH2, which are key intermediates in the ammonia oxidation pathway. The inclusion of these reactions suggests that the LBV would be promoted and could be why Okafor Reduced Mech, lacking these pathways, under-predicts LBV at intermediate xH2.

Other pathways that have a significantly higher IROP in Gotama Mech's reaction pathway are the NH3→NH2→N2H2→NNH→N2 as can be noticed in Fig. 2. From this pathway, the NNH→N2 pathway, which has an IROP 2.2 times greater in Gotama Mech's pathway relative to Okafor Reduced Mech, is facilitated primarily by the reaction NNH=N2+H (97.4%) (R3). This finding suggests that more H radicals are produced by Gotama Mech through this pathway, relative to Okafor Reduced Mech, and correlates with the higher LBV prediction with Gotama Mech.

The prior observation can also be found in the NH3→NH2→NH→N2H2→NNH→N2 pathway. IROPs for these pathways are greater in Gotama Mech's reaction pathway than Okafor Reduced Mech's pathway. Furthermore, another H radical generating reaction from this pathway, specifically the NH2→NH pathway, also suggests strong correlation with increasing LBV. This reaction is NH2+NH=N2H2+H (R4), which also includes NH and NH2, important intermediate species that affect LBV.

(*4) The three observations from Fig. 2 depict the importance N2Hx species and their reaction pathways have on LBV prediction accuracy. Okafor Reduced Mech's LBV under-prediction for NH3/H2/Air flames at intermediate hydrogen concentrations has a correlation with N2Hx reaction pathways omitted or poorly modelled.

This led us to believe that formation and consumption reactions of NH_2 and NH were not accurately represented, resulting in poor flame modelling under these conditions. N_2H_x reaction pathways also displayed involvement in H radical generation which correlated with LBV prediction.

(*4) The comparison between Nakamura Mech and Okafor Reduced Mech reaction pathways was also conducted. Significant differences in IROP of certain pathways with reactions containing NO and this could be linked to the increased concentration of H radicals due to hydrogen co-combustion. More H radicals lead to increased production of O and OH radicals, radicals that activate reaction pathways containing NO. $\text{NH} \rightarrow \text{N}$ pathway has a higher IROP in Nakamura Mech's reaction pathway due to the influx of H radicals. The excess N being produced as a result of this, paired with NO, reaction $\text{N} + \text{NO} = \text{N}_2 + \text{O}$ facilitates the IROP increase in $\text{N} \rightarrow \text{N}_2$ reaction pathway in Nakamura Mech. However, NO-related pathways from Nakamura were not incorporated into Present Mech as insights from Okafor Reduced Mech's LBV of oxygen-enriched combustion [5]. Under such conditions, O and OH radicals are in abundance too however, Okafor Reduced Mech predicted LBV accurately leading us to determine updating the NO-related pathways were of low priority.

2.3 Changes in the New Mechanism

Since N_2H_x reaction pathways are demonstrated to be consequential to LBV modelling by the comparison of Gotama Mech's and Okafor Reduced Mech's reaction pathways for an $\text{NH}_3/\text{H}_2/\text{Air}$ flame, Gotama Mech was used to inform the updates made to Present Mech. Present Mech focuses on better describing N_2H_x -related reaction pathways in the chemical kinetics mechanism. This was achieved by focusing on two aspects of the reaction pathway: N_2H_x reactions in the $\text{NH}_2 \rightarrow \text{NH}$ pathway and N_2H_x formation/consumption to convert NH_x radicals to NNH . Therefore reactions pertaining to species NH_2 , NNH , N_2H_2 and N_2H_4 from Gotama Mech were incorporated into Okafor Reduced Mech to form Present Mech.

[Figure new Mech lbv]

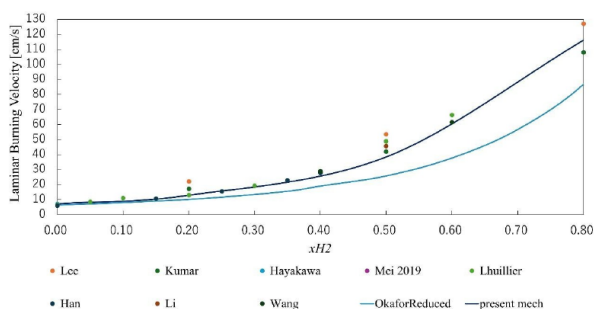


Fig. 3 Experimental and numerically calculated laminar burning velocity (Present Mech and Okafor Reduced Mech) burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ flames under stoichiometric condition at various hydrogen concentrations

Figure 3 compares the LBV predictions of Present Mech to Okafor Reduced Mech and experimental data from literature under the same conditions the reaction

pathway analysis was conducted in. It is evident that the changes made to the Okafor Reduced Mech have been effective to bring calculated LBV for fuel compositions with intermediate and high hydrogen concentrations are closer to real flames' LBV. Further validation will be conducted in this paper to ensure Present Mech fulfils the full scope this study's aim.

3. Analysis of $\text{NH}_3/\text{H}_2/\text{Air}$ Flame Characteristics

3.1 Sensitivity Analysis With Respect to Laminar Burning Velocity

[Figure lbv sens analysis]

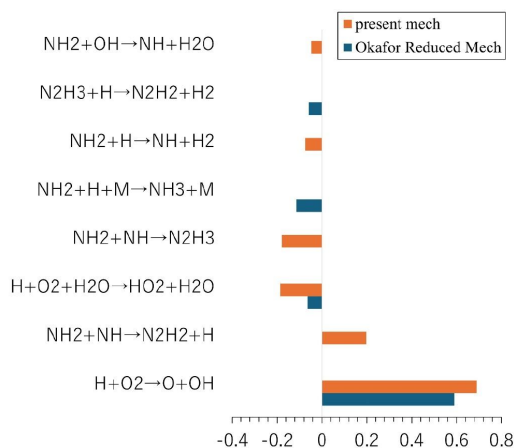


Fig. 4 sensitivity coefficients for laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{Air}$ flames at $x\text{H}_2=0.5$ and equivalence ratio of 1.0 under atmospheric conditions.

From Fig. 4, reaction (R4), $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$, plays a significant role in promoting LBV in Gotama Mech whereas is not significant in Okafor Reduced Mech. Such a pattern can also be noticed in one of the reaction primarily responsible for inhibiting LBV in Gotama Mech, $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_3$ (R1). The figure also shows other NH_x and N_2H_x related reactions being more significant with respect to LBV in Present Mech, suggesting the dominant role these species play in controlling the LBV of an $\text{NH}_3/\text{H}_2/\text{Air}$ flame, at intermediate hydrogen concentration, has been modelled.

3.2 Sensitivity Analysis With Respect to Major Chemical Species

NH_x species, like NH and NH_2 radicals, mole fractions sensitivity to elementary reactions' A-factors are plotted in [fig nhx sens]. Both, Okafor Reduced Mech and Present Mech, agree that the reaction $\text{NH}_3 + \text{OH} \rightarrow \text{NH}_2 + \text{H}_2\text{O}$ (R5) and the reaction $\text{NH}_2 + \text{H} \rightarrow \text{NH} + \text{H}_2$ (R6) are the major reaction involved in NH_3 and NH_2 consumption respectively. However, Present Mech also identifies the reaction $\text{NH}_2 + \text{OH} \rightarrow \text{NH} + \text{H}_2\text{O}$ (R7) significantly contributes to NH_2 consumption as depicted in [fig nhx sens]. Reaction (R7) displays negative sensitivity to NH_2 mole fraction with both mechanisms while exhibiting significant, positive sensitivity with Present Mech to NH mole fraction. Thus, Present Mech promotes NH consumption via reaction (R7). Reaction $\text{NH} + \text{H} \rightarrow \text{N} + \text{H}_2$ (R8), on the other

hand, exhibits significant negative sensitivity to NH mole fraction, suggesting that (R8) is the primary decomposition pathway for NH.

On the other hand, N₂H_x species, as shown in [fig n2hx sens], depend on NH_x radicals to recombine and form NNH, N₂H₂ or N₂H₃. NNH and N₂H₂ mole fraction exhibits dependance on reaction (R4), NH₂+NH=N₂H₂+H, whereas N₂H₃ mole fraction is negatively impacted by the same reaction. The negative sensitivity coefficient for N₂H₃ mole fraction is because N₂H₃ relies on the same reactants, NH and NH₂, in reaction (R1), NH₂+NH=N₂H₃. Since H radical generation is significant via (R1) in NH₃/H₂/Air flames with intermediate xH₂, both reactions are competing for the NH and NH₂ radicals. Reaction (R4) activates the reaction zone whereas reaction (R1) suppresses the reaction zone therefore, both (R1), (R4) and other reactions governing N₂H₃ formation were added to Present Mech ensure the reaction zone is balanced in terms of reactivity. This should in turn improve the LBV prediction accuracy while using Present Mech.

4. Validation

5. Concluding Remarks

References

1. Nakamura, H., Hasegawa, S. & Tezuka, T. Kinetic Modeling of Ammonia/Air Weak Flames in a Micro Flow Reactor with a Controlled Temperature Profile. *Combustion and Flame* **185**, 16–27. ISSN: 0010-2180 (Nov. 1, 2017).
2. Nakamura, H. & Hasegawa, S. Combustion and Ignition Characteristics of Ammonia/Air Mixtures in a Micro Flow Reactor with a Controlled Temperature Profile. *Proceedings of the Combustion Institute* **36**, 4217–4226. ISSN: 1540-7489 (Jan. 1, 2017).
3. Gotama, G. J. *et al.* Measurement of the Laminar Burning Velocity and Kinetics Study of the Importance of the Hydrogen Recovery Mechanism of Ammonia/Hydrogen/Air Premixed Flames. *Combustion and Flame* **236**, 111753. ISSN: 0010-2180 (Feb. 1, 2022).
4. Otomo, J., Koshi, M., Mitsumori, T., Iwasaki, H. & Yamada, K. Chemical Kinetic Modeling of Ammonia Oxidation with Improved Reaction Mechanism for Ammonia/Air and Ammonia/Hydrogen/Air Combustion. *International Journal of Hydrogen Energy* **43**, 3004–3014. ISSN: 0360-3199 (Feb. 1, 2018).
5. Abe, S., Kitagawa, T. & Okafor E. C. A Study on the Elucidation of NO_x Formation Pathways in Oxygen-Enriched Ammonia Combustion. In *Proceedings of the 35th Internal Combustion Engine Symposium [No. 24-51]* 35th Internal Combustion Engine Symposium [No. 24-51] (2024).